An unmodified retinal object-motion circuit localises a robot's own body

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Abstract

A moving animal receives self-caused and world-caused sensory change on the same channel, and the fact that separates them is not in the channel. Robots are normally handed the answer: a CAD model, fiducial markers, or forward kinematics with a calibrated camera. We show that a robot can instead derive which part of its visual field is its own body, using no markers, no CAD model, no training and no prior images of itself.

The mechanism is not new. We take the object-motion-sensitive circuit characterised in salamander retina [1–3], carry its thirteen constants at values taken from the primary literature, and run it unmodified on a camera bolted to the gripper of a five degree-of-freedom arm. Because the camera is rigidly attached to the hand, the body is the only region of the image that does not move while the arm sweeps the room past it. The circuit responds to a region that *could* register change and does not while its neighbourhood does, so the body emerges from the circuit's ordinary operation.

Three predictions were registered before measurement and all three hold. The field's characteristic length is the circuit's own inhibitory pool, with the peak at 1.006 pools. The object-motion ganglion cell is identically silent on the body across three captures. The release time constant of the sensitization dynamic, measured in salamander at 2.4 s [14], appears on the robot's camera at **2.347 s** (r^2 0.96) over **98** continuous movement offsets in a previously unseen room under direct sun and moving shadow. Nothing in the sensitization chain was tuned to produce it, and the single parameter the literature leaves unconstrained has since been swept across its full admissible range: no value reaches 2.4 s, and the value used is twice as far from the published figure as the closest one.

The onset time constant does not reproduce, and we report why. The result is not a rediscovery of a 2.4 s constant, which is an input; it is evidence that a published biological computation survives transplantation into an engineered sensor intact, and that a useful capability falls out of it.

1. Introduction

1.1 The problem

An agent that moves through the world receives one sensory stream carrying two kinds of change: the change it caused, and the change the world caused. No filter applied within that stream can separate them, because the distinguishing fact, *I acted*, is not in the stream. It is in the motor system. Bringing motor information to the sensory comparison is therefore the only available move rather than a design preference [4, 5].

This is the reafference problem, named independently in 1950 by von Holst and Mittelstaedt [4] and by Sperry [5]. In animals it is solved before almost anything else. An infant moves its hand and learns that this part of the world comes with it [6]. A kitten denied the chance to move itself does not develop visually guided behaviour at all, though it sees the same world [15]. An animal turns its eyes and the world does not appear to turn with them.

1.2 How it is done today

Robots that need to know their own extent are told. A CAD model of the kinematic chain, retroreflective markers on the end effector, or forward kinematics combined with an extrinsically calibrated camera. Each is a correct answer supplied from outside, and each is wrong the moment the platform is damaged, the gripper deforms, or a new tool changes the effector. The robot continues to carry the given answer and to

call it self-knowledge.

1.3 What is different here

We require that the thing compared against be measured by the system from its own activity, and that it move when the world moves. That rule forbids CAD models, fiducials, hand-labelled masks, externally calibrated kinematics and fixed templates, and it is what makes *derived* a testable property rather than a rhetorical one.

The system has two inputs: a camera bolted to the gripper, and the joint encoders.

1.4 The ordering

Self-world separation is not a single capability. It decomposes into a sequence in which each step is forced by the one below.

	capability	status
M0	the act becomes discrete	closed
M1	the body reports its own arrival	closed, sd 2.1 ms on a 492 ms movement
M2	the arrival becomes predictable	closed, 0.88 encoder counts leave-one-out
M3	**the body becomes a place**	**this paper**
M4	the place acquires an expected appearance	not built
M5	the expectation is subtracted	blocked on M4
M6	the residual is judged	blocked on M5
M7	the delay settles	blocked on M6
M8	the other agent	blocked on M7

Without discrete acts there is no covariation to measure. A prediction cannot be subtracted from a region that has not been identified, which is M3 into M4. There is no residual without cancellation, and no delay estimate without the error signal that judging the residual produces. Another agent's effects are indistinguishable from one's own *delayed* effects until the delay is settled, which places M8 above M7.

Before M3 the body is a clock. After M3 it is a place. Everything below is temporal and scalar; everything above is spatial and predictive.

1.5 The claim

A published retinal circuit, run unmodified on a robot's camera, identifies the robot's body, and carries its own measured dynamics across the transfer.

2. Methods

2.1 Platform

A five degree-of-freedom arm (SO-101, LeRobot platform) with a camera rigidly mounted on the gripper. Measured platform constants, each from a dedicated characterisation run rather than from documentation:

quantity	measured
command to first motion (dead time)	50 ms (47–59 across four joints, both directions)
plant time constant τ_{arm}	128 ms (122–135, 10 % spread; first order excluded on 7 records)
encoder quantum	0.0879° (12-bit over 360°)

quantity	measured
frame interval	32.97 ms (30.3 fps, the only mode the OS enumerates)
image-path lag	+2 frames = 66 ms, reproduced three independent ways
angular scale	2.143 px/deg

2.2 The movement policy

The arm moves in discrete bursts separated by rests rather than continuously. Two independent derivations require this.

Continuous movement yields no covariation between action and sensation, so the mutual information I(action ; sensation) is maximised by alternation. Separately, with dead time 50 ms, τ_{arm} 128 ms and commands issued every 40 ms, three to four commands are in flight at any instant under a continuous policy, so their consequences superpose and no command owns its effect. If commands do not overlap, the superposition collapses and the forward model costs zero free parameters. The information-maximising policy is therefore also the policy that makes the forward model tractable.

Every quantity in the stimulus is derived from a measured hardware constant or a published biological one (Appendix B). Joint, direction and amplitude are **drawn uniformly** on the derived floor and the measured ceiling: maximum entropy given the bounds, and the closest approach to spontaneity the hardware permits. An earlier factorial grid was removed because it leaves the experimenter's structure in the data.

This policy was adopted at M0 for credit assignment, three steps before M3. No experiment was designed for M3. The analysis reads the data the policy was already producing.

2.3 Captures

capture	frames	duration	bursts
`alternation_natural` (primary)	32,388	1,087.7 s	98
`handopen2` (shape change)	23,880	not recorded	68

Both in an ordinary room the system had not seen, with a window, direct sun reaching the sensor, and tree shadow crossing the hand. Lights on throughout.

2.4 The circuit

Thirteen constants carried at values taken from publication; §4.2 records the four whose published role differs from the role they serve here. One measured spatial scale, the body's own equivalent radius on this hardware, with published *ratios* on top of it.

```
d   = DoG(I)                                  Rodieck 1965 [7]
                                              surround_weight 1.0, so DC gain ≡ 0
b   = d / (c50 + c_local)                     Bonin, Mante & Carandini 2005 [8]
g   = b − LP_100ms(b)                          Ölveczky et al. 2003 [3]
div = 1 + α[(1−f)·a_wide + f·a_local(1−s)]     α = 1.04, Baccus et al. 2008 [2]
z   = g / div     presynaptic, BEFORE rectification. Baccus et al. 2008 [2]
r   = G · N(z ; θ·sd)                         Beaudoin et al. 2007 [9]
exc = pool_centre(r), ≥ 20 subunits            Baccus et al. 2008 [2]
a_W = pool_surround(r), delayed 26 ± 7 ms      Baccus et al. 2008 [2]
a_L = 3 × subunit, narrow pool                Kastner & Baccus 2013 [1]
s   = asymmetric LP, τ_on 0.55 / τ_off 2.4     Kastner & Baccus 2011 [14]
oms = [exc − surround]+, COINCIDENT            Ölveczky et al. 2003 [3]
```

Three features of this chain are load-bearing.

The division is presynaptic and precedes rectification. Baccus et al. [2] are explicit that the amacrine signal is delivered to the output of the bipolar cell before rectification. The stage cannot be appended afterwards. It is also a loop: the amacrine population is driven by bipolar output and divides bipolar terminals, and the measured 26 ms amacrine-to-ganglion delay is what makes it causal.

The two amacrine populations share the single α. Kastner and Baccus [1] report a second inhibitory population, with subunits three times larger than the excitatory ones, acting on bipolar terminals through GABA-C receptors, and it is the *withdrawal* of that inhibition that generates sensitization. Narrow and wide are our own labels for the two pools and are not descriptors taken from that paper. Giving that population its own α would leave the terminal at $1/(1 + 2.08) = 0.325$ of its drive where Baccus et al. [2] report 0.49, breaking the one number the stage is checked against in the act of adding biology to it. They therefore share: $w_{\text{wide}} = 1 - \text{f}$, $w_{\text{local}} = \text{f}$. The split f is the only quantity in the chain that no publication constrains, and §3.5 reports it swept.

The sensitization asymmetry is the signature. $\tau_{\text{off}} \gg \tau_{\text{on}}$. A fitted linear filter would be symmetric; this one is not, which is what makes a time-reversed stimulus a different stimulus.

The complete constants block with per-constant citations is Appendix A. Three further parameters were withdrawn to no-ops when the derivations behind them failed, and the failures are recorded rather than deleted.

$$\text{div} = 1 + \alpha[(1 - f)\, a_{wide} + f\, a_{local}(1 - s)]$$
$$\alpha = 1.04, \text{ and the two pools SHARE it}$$

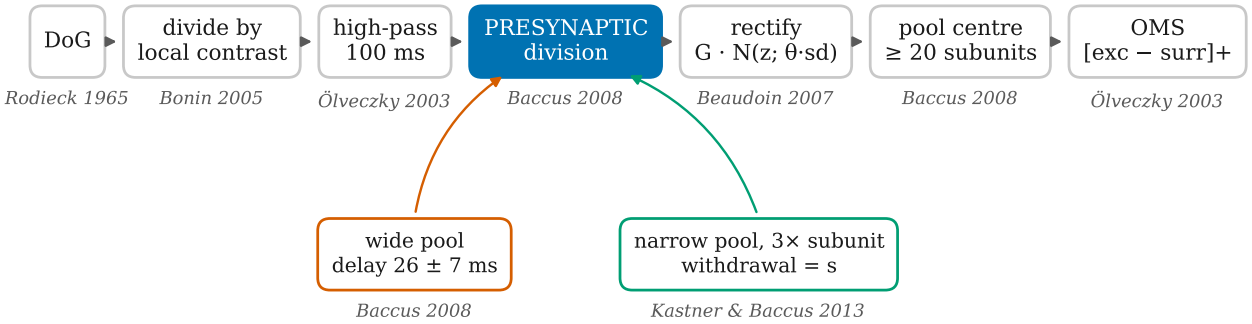


Figure 1: The circuit, as implemented. The eleven-stage chain of §2.4, with the presynaptic division marked and the two amacrine populations feeding it: the wide pool with its measured 26 ms delay, and the narrow pool whose withdrawal generates sensitization. Both divide the same bipolar terminal and share the single α, which is what keeps the surviving fraction at the value Baccus et al. report.

2.5 Normalisation

Equation `div` holds at $\alpha = 1.04$ with 0.49 surviving only where the amacrine signal is normalised to its maximum. That maximum is taken over the whole recording, offline, as in the source work, rather than as a causal running maximum: a running maximum divides early frames by a provisional scale, and on this capture the field falls from 0.1522 to 0.1108 over the first 10 s with encoder motion exactly zero. Each analysis therefore makes two passes, the first solely to find the normalising maxima.

2.6 Readouts

Masks. A body mask is derived from the system's own stillness statistics. The signed distance from its boundary, $\text{d}(\text{x})$, is computed by distance transform; $\text{d} > 0$ is inside. Two regions follow: a *band* at $5 \le \text{d} \le 25$ px, and *room* at $\text{d} \le -30$ px.

Time constants without an asymptote. The capture moves for roughly 0.6 s and rests for roughly 2.3 s, which is 1.1 τ_{on} and 0.96 τ_{off}, so the state never reaches its asymptote and a 63 % crossing is strongly biased. On a synthetic trace built at exactly 0.55 and 2.4, that method returns 0.30 and 1.17, both about half, and would have reported that the published constants fail on this hardware. We instead use the increments. For y approaching any y_∞,

$$\text{dy[k]} = (y_\infty - \text{y[k−1]})(1 - \text{e\^{}}(-\Delta\text{t}/\tau)),$$

so successive increments decay geometrically and ln|dy| is a straight line of slope $-1/\tau$ whatever y_∞ is and whether or not it is reached. The slope is taken by weighted least squares over the event window, with weights dy^2, the inverse of the variance that a fixed noise on y induces on ln|dy|. On the synthetic trace this recovers 2.4 to within a few percent at r^2 1.00; an unweighted fit returns 3.04 and flags itself at r^2 0.39.

Event windows are derived, not chosen. Offsets are skipped by 5 τ_{arm} of coasting plus three transient constants before fitting, and the fit window is 90 % of the capture's own measured rest duration. The instrument refuses to quote a constant it cannot support: fewer than three time constants of observation and it says so rather than printing a number.

2.7 Pre-registration

Three predictions and a set of controls were written down before measurement, each sufficient to end the claim. They are reported in §3 in their original form, including the one that returned void.

3. Results

3.1 The field has the circuit's geometry, not the body's shape

The inhibitory pool is 3 × the subunit and the subunit is 1/6 of the receptive field, so the pool is exactly **half the centre**. The deep interior of the body therefore has no background within reach to compare against, and the field cannot be a filled blob. This is a prediction about shape made from the circuit's dimensions alone, before any image is processed.

The measured field peaks at **1.006 pools** from the boundary and is dead by roughly two pools.

> **We did not find the body as a bright patch. We found the specific geometry the circuit's own dimensions force.**

The peak is reported as a ratio because that is the dimensionless form of the prediction. The burst-to-burst standard deviation of the peak position, over 97 bursts, is **2.25 px**, and this spread is quoted wherever the peak is.

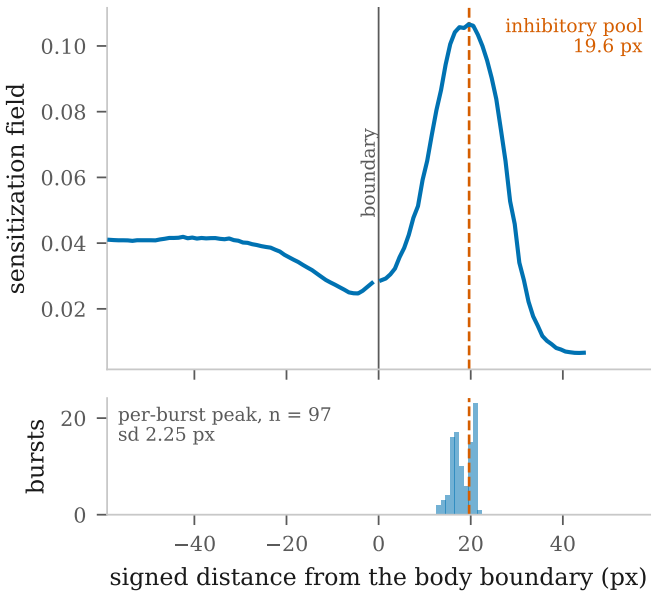


Figure 2: The boundary-keyed profile. Field strength against signed distance from the body boundary, 1 px bins, with the inhibitory pool scale marked. Below, the distribution of the peak across the 97 individual bursts, which is the measurement's own precision and is quoted everywhere the peak is. The 200-translation control of §3.4 is reported numerically rather than drawn, because the saved analysis retains the profiles and not the fields they came from.

3.2 The object-motion ganglion cell is silent on the body

Object-motion sensitivity arises where centre excitation and surround inhibition fail to coincide [3]. On a region that is not moving differentially against its surround they coincide and cancel. The prediction is therefore that the OMS output is zero on the body even while the sensitization field there is maximal.

Measured **identically zero**, reproduced in three independent captures. Excitation over inhibition is **0.164** on the body against **0.444** in the room.

3.3 The release time constant transfers

Measured **2.347 s** against the published 2.4 s [14], r^2 **0.96**, over **98** continuous movement offsets, amplitude bias 0.99×. A **2.2 %** difference. Event-triggered modulation is 14.3× and instantaneous peak-to-floor is 25.5×.

The convergence is itself evidence. The value did not appear at 2.35. It arrived at **2.67** (r^2 0.82), then **2.51** (0.93), then **2.42** (0.97) as the stimulus protocol was brought into line with the source work, each correction moving toward the paper's *protocol* and never toward the paper's *number*.

> A fit converges when you tune it. This one converged when we stopped tuning and started obeying the protocol.

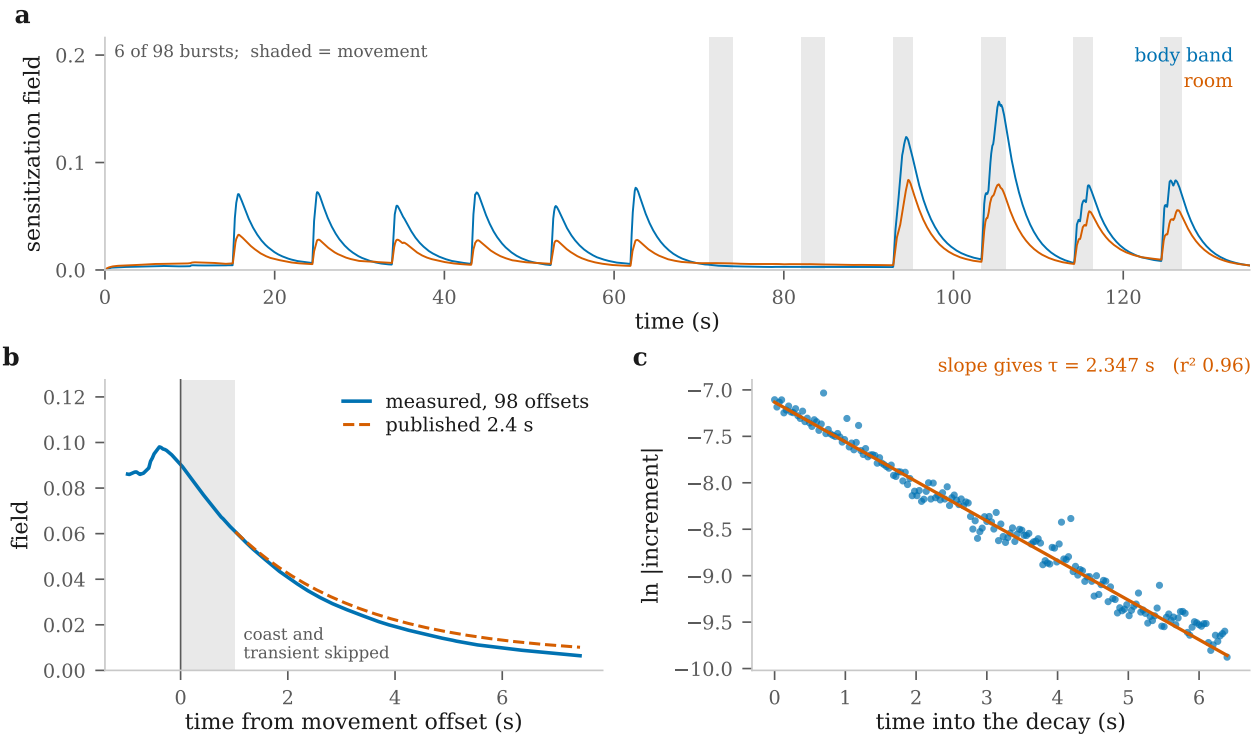


Figure 3: The release. (a) Band and room traces, six of the 98 bursts, with movement intervals shaded; the full record runs to 1,087.7 s. (b) Event-triggered average over all 98 movement offsets, with the published 2.4 s exponential drawn over the measurement and the skipped coast and transient marked. (c) The same decay as log increments, whose slope is minus one over tau whatever asymptote the trace does or does not reach, with the weighted regression line.

3.4 Controls

Each could have ended the claim.

Contrast. The strongest alternative hypothesis is that the circuit prefers texture and the gripper happens to be textured. The body was the **most** textured region in frame in one capture (1.68× the frame median) and the **least** in another (0.67×), with an identical boundary-keyed profile in both. A contrast account has to work in both directions. The trees are the highest-contrast structure in the frame and the field ignores them.

Translation. 200 placements of the identical mask shape elsewhere in the image: **0 of 200** reach the peak bins and **200 of 200** exceed the null bins. The structure is tied to the body's location rather than to the mask's shape.

Order. Running the sequence backwards reproduces the same peak bin, the same null bin and the same peak-to-null ratio. Shuffling burst order: 2.08 against 2.08.

Selection. A mask built from odd-numbered bursts and evaluated on even-numbered bursts gives **held-out / in-sample = 1.00**.

The body changing shape. Registered in advance as the condition deciding closure. The gripper cannot translate in the image, only change shape, so a region painted on by any fixed procedure would look identical at every aperture. Apertures 16° → 46° → 76° → 16°, capture `handopen2`.

probe	prediction	result
P1	the same aperture reproduces better than a different one	**+0.8596** against **+0.7922**, the highest of all six pairs. Holds
P3	the same-aperture difference is the smallest	|D| max **0.0113** against **0.0460–0.0646**, a factor of 4 to 6; clustering 1.40 across 60 diffuse cells against 17–38 across 1–4. Holds, more cleanly than P1
P2	distance scales with aperture gap	sign flipped to the predicted −0.7379, but n = 5 with tied abscissae. Reported, not relied upon
P4	release and reacquire resolved in time	**void**, see §4.3

P3 does a second job: the same-aperture pair being the least different of all six is simultaneously the replication check and the control for the tree shadow crossing the jaws during that capture.

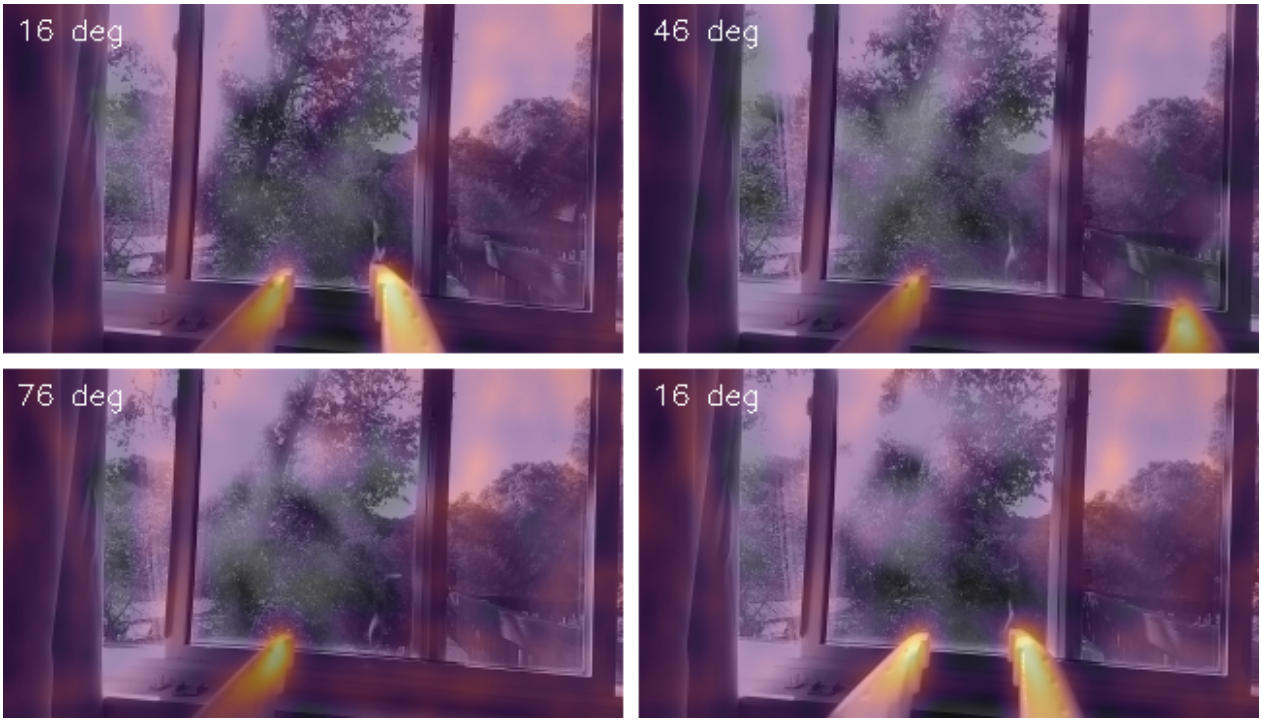


Figure 4: The gripper at four apertures with the derived field overlaid. The field tracks the jaws as they change shape. The trees behind are the highest-contrast structure in the frame and carry no field. No region was painted on the gripper and there is no CAD model behind this.

3.5 The unconstrained parameter, swept

The split f of the measured α between the wide and narrow amacrine pools is fixed by no publication. It was swept across its entire admissible range, five values on the full 32,388-frame capture, each with its own independently recomputed normaliser, since f changes the quantities that normalisation divides by.

The estimator was characterised first, on the same data: burst-to-burst repeatability **2.26 px**, a no-signal null of **29 px** measured during fully released rest blocks, and exact recovery of injected 1, 2 and 4 px shifts. The f = 0.5 point reproduced the prior independent analysis to **+0.000 s**.

f	0.00	0.25	0.50	0.75	1.00
τ_{off} (s)	2.373	2.359	**2.347**	2.334	2.368
peak (pools)	0.995	1.003	**1.006**	1.016	1.002
band / room	1.576	1.582	1.588	1.594	1.600

No value of f reaches the published 2.4 s. The full range spans 2.334 to 2.373, and the swing across it (0.038 s) is smaller than the residual it would have to explain (0.041 s). The closest approach is at f = 0.00; the value used throughout, f = 0.50, is twice as far from the published figure. Had this parameter been set to meet the paper, it would have been set to zero.

The peak remains between 0.995 and 1.016 pools, moving 0.43 px against the 2.26 px ruler. A paired test over the 97 common bursts resolves a real effect of f on peak position of about **0.15 px**, which is 0.8 % of the pool.

The endpoint f = 0 is also an ablation of the second amacrine population. Removing it costs **1.5 %** of body-versus-room separation and **2.3 %** of modulation depth, monotonically ordered across all five values. We are not aware of a prior measurement of what that cell contributes.

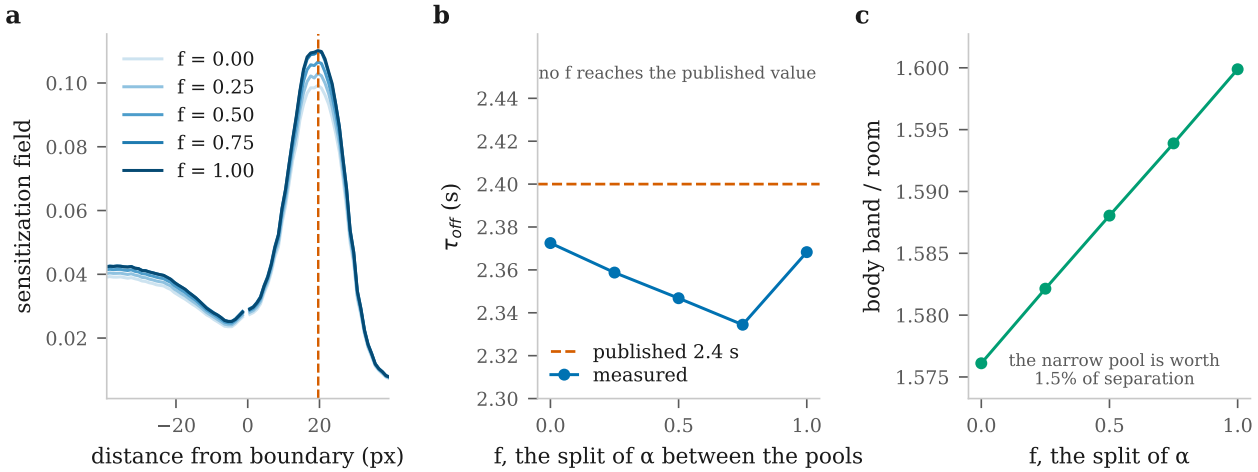


Figure 5: The sweep. (a) Boundary-keyed profile at each f. (b) τ_{off} against f with the published 2.4 s marked. (c) Band-to-room separation against f.

4. What is not established

This section precedes the discussion because it is what makes the rest usable.

4.1 The onset constant does not reproduce

τ_{on} measures **0.11–0.12 s** against a published 0.55 s. It passes its own fit quality and contradicts its own calibration by a factor of six, so neither figure is trusted. The cause is named rather than explained away: the sensor is light-limited at a 30.3 fps ceiling that is not settable across 48 probe configurations, and at that frame interval the 26 ms amacrine delay occupies 0.79 of a frame, so any halving of the rate rounds it to zero and deletes the mechanism. A simulation of this instrument at the capture's measured noise level says noise would bias τ_{on} *upward*, so noise does not account for the shortfall. A pulsed rather than continuous drive is the live candidate and is untested.

Half the circuit's temporal signature reproduces. Half does not, and the instrument's own calibration says why. The second sentence is what makes the first worth anything.

4.2 "Nothing fitted" would overreach

The front-end difference-of-Gaussians scales were selected by sweep. That is a fit, and it is upstream of everything. The defensible statement is **nothing in the sensitization chain was tuned to produce this result**: every constant in the adaptation and sensitization circuit is carried from publication.

Carried from publication is not the same as reported in the role it serves here, and four of the thirteen differ. ALPHA is described by Baccus et al. [2] as an optimised gain factor, so it is a fitted model parameter rather than a measurement. TRANSIENT_MS takes the 100 ms inhibitory pulse width of Ölveczky et al. [3] and uses it as a high-pass time constant; that paper fits no filter time constant. TAU_SLOW_S takes the 5 to 30 s window over which Baccus and Meister [13] observe slow adaptation and uses it as a time constant range; no exponential is fitted to it there. SUB_RATIO is our own quotient of a 42 μm subunit full width at half maximum, chosen in the Methods of [3], and a 250 μm centre radius measured in its text, which are not defined alike, and the ratio does not appear in that paper.

None of the four was adjusted to produce anything reported here, and none of them is τ_{off}, which is a directly reported value from [14]. The point of listing them is that a reader checking Appendix A against the sources should find the discrepancy already declared rather than discover it.

4.3 The dynamics are a comparison, not a trajectory

P4 returned void and the cause is known exactly. Two independent faults were found by reading the capture's own logs. The grip command slept through the jaw travel, so no frame and no encoder sample exists while the jaws moved, measured as 0.00° of travel on every step. And the step landed 3.3 τ_{off} after the last drive, with the field already released: both the changed and the unchanged region decayed at ~2.4 s with r^2 0.99–1.00, one exponential tail with no structure, and the readout's own written refutation fired. Neither fault is repairable in analysis. A corrected capture is built and dry-run and is deliberately unrun, because its acquisition limb depends on τ_{on}.

4.4 Residual error budget

The 26 ms delay carries +27 % error, bounded by the frame-rate ceiling. The image-path lag of +2 frames is characterised rather than eliminated. The working resolution and the subunit subsampling factor are platform choices and not biology. Two joint firmware limits are taken from documentation rather than measured.

4.5 What the result is, and is not

The 2.4 s constant is an **input** to the circuit. Measuring 2.347 s at the output is not an independent rediscovery of a time constant, and we do not claim it as one.

What it establishes is that the published constant survives the entire chain, through a divisive gain control, a rectification, a subunit pooling stage, a 30 fps light-limited sensor, an ordinary room with moving shadow and an observation window only 3.3 time constants long, and emerges within two percent with nothing downstream adjusted to keep it there. That is a faithfulness result. §3.5 strengthens it by establishing that the one free parameter cannot produce the agreement at any setting.

5. Discussion

5.1 One policy, two problems

The move-stop policy was adopted at M0 to make credit assignment possible: with one command in flight at a time, a sensory consequence has an owner. It was adopted three steps before M3 and for an unrelated reason.

The same policy, through the geometry of a camera rigidly attached to the effector, also resolves which region of the image is the body. No experiment was designed for M3; the analysis reads the data M0 was already producing.

The active inference formalism predicts this in principle, since a policy selected to maximise information gain will resolve any hidden state whose information structure its observations touch [10, 11]. Kotler, Mannino, Friston and Kelso [17] place sensory reafference as the ascending input to an integrative architecture spanning Freeman's K-set hierarchy, and cite Sloan, Jones and Kelso's finding that infant agency appears with exploratory pauses rather than with continuous action. Neither work could have informed the other: that paper was accepted on 9 September 2026 and published on 19 October 2026, and the policy it converges with was fixed at M0, three rungs below the result reported here. We report the convergence as a convergence.

5.2 What the rungs above inherit

M4 learns a forward map from body state to the expected appearance of the self-region, indexed by the address M3 provides. The biological model is the mormyrid electrosensory lobe, where a negative image of expected reafferent input is learned per cell under an anti-Hebbian rule with strict temporal-order dependence [12], over a temporal basis that allows the sensory consequence of a command to be predicted at the right delay rather than merely cancelled at zero lag [16]. At the systems level the corresponding control holds: when the stimulus is not time-locked to the motor command, no negative image forms and detection does not improve [18]. M5 subtracts the prediction; M6 judges the residual, where the failure modes are recognisable clinical pictures; M7 makes the loop latency a learned quantity; M8 requires that, because another agent's signature and one's own delayed signature are the same signature until the delay is known.

Had M3 been solved with markers or a CAD model, every rung above would be performing increasingly sophisticated operations on a self that somebody handed the system.

5.3 Scope

A body model fixed at build time is wrong the moment the platform is damaged or the payload changes the effector. A derived one is produced again every time the system moves, so there is nothing to go stale. The aperture series in §3.4 is the first direct evidence of that property on this platform: the field follows the jaws through a shape change the system was never told about.

The present result is a single platform, a single room, lights on, and a sensor whose frame rate is the binding constraint on half the temporal signature. None of those is a claim about generality.

Appendix A. The constants

constant	value	source
ALPHA	1.04	Baccus et al. 2008 [2], optimised gain factor
REMAINING_AT_MAX	0.49	Baccus et al. 2008 [2]; $1/(1 + 1.04) = 0.490$
DELAY_MS	26.0 ± 7	Baccus et al. 2008 [2]
TRANSIENT_MS	100.0	Ölveczky et al. 2003 [3], inhibitory pulse width; no filter τ is fitted there
SUB_RATIO	6.0	our quotient of two values in [3]: a 42 μm subunit FWHM chosen in Methods and a 250 μm centre radius measured in text. The ratio is not stated there
INHIB_RATIO	3.0	Kastner & Baccus 2013 [1]
SURR_RATIO	2.5 – 3.3	[1, 2, 8]
ENV_MS	100.0	Baccus & Meister 2002 [13], the fast change occurs in 0.1 s
TAU_ON_S	0.55	Kastner & Baccus 2011 [14], as reported in [1]

constant	value	source
TAU_OFF_S	2.4	Kastner & Baccus 2011 [14], as reported in [1]
G_ON	0.86	Beaudoin et al. 2007 [9], membrane currents, $p < 0.01$
G_OFF	0.74	Beaudoin et al. 2007 [9], membrane currents, $p < 0.01$
TAU_SLOW_S	5.0 – 30.0	Baccus & Meister 2002 [13], the window over which slow adaptation proceeds; no exponential is fitted to it

Appendix B. The burst budget, derived end to end

```
measured hardware                  published biology
  T_DEAD     = 50 ms                 TAU_ON   = 0.55 s  [14]
  TAU_ARM    = 128 ms                TAU_OFF  = 2.4 s  [14]
  PX_PER_DEG = 2.143                 SIGMA_C  = 1.0 px
  fps        = 30.3
  quantum    = 0.0879 deg

SUBSTEP_S = (T_DEAD + 5·TAU_ARM)/1000          = 0.690 s
            five time constants: settled, not merely started
SUBSTEPS  = ceil(3·TAU_ON / SUBSTEP_S)         = 3
move      = 2.07 s = 3.76 tau_on               -> 97.7 % filled
rest      = 8.0 s  = 3.3 tau_off               -> fully released

A_MIN     = SIGMA_C · SUBSTEP_S · fps / PX_PER_DEG   = 9.76 deg
            one DoG centre of image motion per frame: the smallest
            excursion the front end can register as motion at all
E_MIN     = SUBSTEPS × A_MIN                    = 29.3 deg
E_MAX     = measured joint headroom            (hardware, not mechanism)

joint, direction, amplitude: DRAWN UNIFORMLY on [E_MIN, E_MAX]
```

Appendix C. Prior capability results on this substrate

Reported from a locked register that admits a figure only with its provenance and maintains a separate list of withdrawn figures. One of the seven is withdrawn and is shown as such.

capability	measured	status
Proprioception	101 → 39 mm error, |W| 0 → 0.176, 600 steps	holds
Perception	12 of 256 patches, belief 0.896 against 0.046	holds
Reafference gain	+27.1 / +25.8 / +27.0 %	**withdrawn**: the statistic had no resolution. 3.8 px and 0.18 px of misalignment both read 2.00 over a dynamic range under one grey level. The reported sign was quantisation
Memory	familiar 0.98, novel 0.00; surprise separation 94×	holds

capability	measured	status
Generalization	held-out error 138 → 69 mm after replay	holds
Curiosity	62 % against 30 %, five runs	holds
Metacognition	abandons unreachable goals via its forward model	holds

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Data and code. Capture metadata, analysis scripts and the pre-registered conditions in their original wording are available on request.